Copyrighted Materials
Copyright © 2005 AIAA Retrieved from www.knovel.com

Table of Contents

Preface ..... **xiii**

I. Overview

Chapter 1. Combustion Instabilities: Basic Concepts **3**
Ben T. Zinn and Timothy C. Lieuwen, *Georgia Institute of Technology, Atlanta, Georgia*

Introduction 3
Historical Overview 5
Causes of Instabilities 8
Growth and Saturation of Instabilities 16
Conclusion 22
References 24

II. Case Studies

Chapter 2. Combustion Instabilities in Industrial Gas Turbines: Solar Turbines' Experience **29**
Kenneth O. Smith and James Blust, *Solar Turbines, Inc., San Diego, California*

Introduction 29
Lean Premixed Combustion System Configurations and Operating Conditions 30
Commercial Introduction at 42 ppmv NOx 33
Emissions Reduction to 25 ppmv NOx 35
Combustor Pressure Oscillation Model 37
Centaur CPO Reduction 37
Mars CPO Reduction 38
Recent Experience 40
Conclusion: Needs and Future Challenges 40
References 41

Chapter 3. Incorporation of Combustion Instability Issues into Design Process: GE Aeroderivative and Aero Engines Experience ... 43

H. C. Mongia, T. J. Held, G. C. Hsiao, and R. P. Pandalai, *GE Transportation, Cincinnati, Ohio*

Introduction ... 43
Fundamental Causes of Combustion Dynamics ... 45
Control Strategies ... 45
Examples of Combustion Dynamics ... 46
Combustion–Acoustic Modeling ... 51
Acoustic Modeling Results ... 57
Conclusion ... 61
References ... 61

Chapter 4. Combustion Instability and Its Passive Control: Rolls-Royce Aeroderivative Engine Experience ... 65

Tomas Scarinci, *Rolls-Royce Canada, Quebec, Canada*

Overview of the Trent 60 Aeroderivative ... 66
Oscillatory Combustion in the Trent 60 DLE ... 68
Combustion System Design Modifications ... 87
Acknowledgments ... 87
References ... 88

Chapter 5. Thermoacoustic Design Tools and Passive Control: Siemens Power Generation Approaches ... 89

Werner Krebs, Sven Bethke, Joachim Lepers, Patrick Flohr, and Bernd Prade, *Siemens AG, Mülheim, Germany* and Cliff Johnson and Stan Sattinger, *Siemens AG, Orlando, Florida*

Introduction ... 89
Siemens Gas-Turbine Products ... 89
Phenomenological Description ... 90
Solution Methods ... 93
Application ... 100
Conclusion ... 110
References ... 111

Chapter 6. Characterization and Control of Aeroengine Combustion Instability: Pratt & Whitney and NASA Experience ... 113

Jeffrey M. Cohen and William Proscia, *Pratt & Whitney, East Hartford, Connecticut* and John DeLaat, *NASA Glenn Research Center, Cleveland, Ohio*

Introduction ... 113
Engine Combustion Instability ... 115
Engine Acoustic Analysis ... 116
Fuel Injector–Air Swirler Dynamic Response ... 118
Subscale Combustor Experiment ... 127

Active-Control Demonstration 135
Conclusion 143
Acknowledgments 144
References 144

Chapter 7. Monitoring of Combustion Instabilities: Calpine's Experience 147

Jesse B. Sewell and Peter A. Sobieski, *Calpine Turbine Maintenance Group, Pasadena, Texas*

Introduction 147
Combustion-Dynamics Monitoring System 148
General Instability Characteristics and Tuning Considerations 151
Detrimental Impacts of Combustion Dynamics 153
CDM for Combustor Health Monitoring: Case Studies 154
References 162

Chapter 8. Monitoring Combustion Instabilities: E.ON UK's Experience 163

Catherine J. Goy, Stuart R. James and Suzanne Rea, *E.ON UK, England, Nottingham, United Kingdom*

Introduction 163
Why Monitor Combustion Dynamics? 163
Description of the On-Line Combustion-Monitoring System 164
Benefits of Combustion-Dynamics Monitoring 165
Case Studies 166
Impact of Ambient Conditions on Dynamic Response 166
Impact of Operating Regime on Dynamic Response 167
Combustion Liner Failure 168
Burner Assembly Failure 171
Conclusion 175
References 175

III. Fundamental Processes and Mechanisms

Chapter 9. Combustion Instability Mechanisms in Premixed Combustors 179

Sébastien Ducruix, Thierry Schuller, Daniel Durox, and Sébastien Candel, *CNRS and Ecole Centrale Paris, Châtenay-Malabry, France*

Introduction 179
Acoustics for Reacting Flows 183
Heat Release as a Pressure Source 187
Heat-Release Fluctuations Driven by Waves 197
Conclusion 206
References 207

Chapter 10. Flow and Flame Dynamics of Lean Premixed Swirl Injectors 213

Ying Huang, Shanwu Wang, and Vigor Yang, *Pennsylvania State University, University Park, Pennsylvania*

Introduction 213
Cold Flow Characteristics of Swirl Injectors 216
Flame Dynamics of Axial-Entry Swirl Injector 250
Conclusion 269
Acknowledgments 270
References 270

Chapter 11. Acoustic-Vortex-Flame Interactions in Gas Turbines 277

Suresh Menon, *Georgia Institute of Technology, Atlanta, Georgia*

Introduction 277
Length and Time Scales 278
Theoretical Considerations 280
Factors Affecting AVF Interactions 287
Conclusion 310
Acknowledgments 310
References 310

Chapter 12. Physics of Premixed Combustion-Acoustic Wave Interactions 315

Timothy C. Lieuwen, *Georgia Institute of Technology, Atlanta, Georgia*

Introduction 317
Background 318
Heat-Release Response to Flow and Mixture Perturbations 323
Conclusion 361
Acknowledgments 362
References 362

IV. Modeling and Diagnostics

Chapter 13. Acoustic Analysis of Gas-Turbine Combustors 369

Ann P. Dowling and Simon R. Stow, *University of Cambridge, Cambridge, England, United Kingdom*

Introduction 369
Linearized Equations of Motion 371
One-Dimensional Disturbances 374
Modal Solutions 389
Application to Gas-Turbine Combustors 394
Modal Coupling 401
Acoustic Absorbers 403

Limit-Cycle Prediction 406
Conclusion 409
Appendix: Derivation of Eq. (13.41) 410
References 411

Chapter 14. Three-Dimensional Linear Stability Analysis of Gas Turbine Combustion Dynamics 415

Danning You, Vigor Yang, and Xiaofeng Sun, *Pennsylvania State University, University Park, Pennsylvania*

Introduction 417
Theoretical Formulation 418
Solution Procedure 433
Sample Studies 433
Conclusion 442
Acknowledgments 442
References 442

Chapter 15. Implementation of Instability Prediction in Design: ALSTOM Approaches 445

Christian Oliver Paschereit, *Hermann-Föttinger-Institute, Berlin University of Technology*, *Berlin*, *Germany* and Bruno Schuermans, Valter Bellucci, and Peter Flohr, *ALSTOM Power Ltd, Baden, Switzerland*

Introduction 445
Network Representation of Thermoacoustic Systems 447
Experimental Determination of Transfer Matrices and Source Terms 449
Modeling the Burner Transfer Matrix 454
Reduced-Order Modeling of Complex Thermoacoustic Systems 461
Application to a Gas-Turbine Combustor 475
Conclusion 478
References 479

Chapter 16. Experimental Diagnostics of Combustion Instabilities ... 481

Jong Guen Lee and Domenic A. Santavicca, *Pennsylvania State University, University Park, Pennsylvania*

Introduction 481
Pressure Measurements 482
Chemiluminescence Measurements 485
Infrared-Absorption Measurements 500
Laser-Induced Fluorescence Measurements 506
Laser Mie Scattering 517
Phase Doppler Particle Analysis 521
Conclusion 522
Acknowledgments 524
References 524

V. Combustion Instability Control

Chapter 17. Passive Control of Combustion Instabilities in Stationary Gas Turbines 533

Geo A. Richards and Douglas L. Straub, *U.S. Department of Energy, Morgantown, West Virginia* and Edward H. Robey, *Parsons Project Services, Morgantown, West Virginia*

Introduction 534
Control-System Models 534
Methods to Improve Combustion Stability 550
Acoustic Dampers 560
Conclusion 569
Acknowledgments 570
Appendix 570
References 575

Chapter 18. Factors Affecting the Control of Unstable Combustors 581

Jeffrey M. Cohen, *Pratt & Whitney, East Hartford, Connecticut* and Andrzej Banaszuk, *United Technologies Research Center, East Hartford, Connecticut*

Introduction 582
Description of the Combustor 585
Actuated Fuel Mixing 588
Actuation Time Delay 594
Fundamental Limitations of Achievable Performance 601
Conclusion 606
Acknowledgments 606
References 607

Chapter 19. Implementation of Active Control in a Full-Scale Gas-Turbine Combustor 611

Jakob Hermann, *IfTA GmbH, Groebenzell, Germany* and Stefan Hoffmann, *Siemens AG, Mülheim, Germany*

Introduction 611
Implementation of AIC on Siemens-Type Vx4.3A Land-Based Gas Turbines 613
Results and Experiences with AIC during Gas-Turbine Operation 620
AIC Fault Tolerance and Long-Term Experiences 631
Advantages of Active Measures 632
Conclusion 632
References 633

Subject Index 635

Author Index 657

Supporting Materials 659